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Causal Sufficient Dimension Reduction for Multiple Continuous Exposures

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MOTIVATION

Environmental mixture studies ask how health outcomes change under exposure to multiple pollutants. The National Institute of Environmental Health Sciences (NIEHS) is interested in understanding the *joint effect of exposures* instead of treating pollutants in isolation.

Current strategies are not causal and are restricted to the linear additive confounding model. Suppose Y is an outcome, X are the exposures, and C are confounders:

$$Y = h(X) + \gamma^T C + \varepsilon, \quad X \in \mathbb{R}^p.$$

Methods differ in how they handle the true latent exposure dimension d and complexity of $h(\cdot)$, creating tradeoffs between the flexibility, interpretability, and causal validity of the estimated relationship between exposure and outcome.

Method Class	Flexible	Interpretable	Supervised	Causal
Full ERS ($d = p$)	✓	×	✓	×
Single-index ($d = 1$)	×	✓	✓	×
Dimension reduction ($1 \leq d \leq p$)	✓	✓	×	×
CSDR (ours)	✓	✓	✓	✓

GOAL: Use sufficient dimension reduction (SDR) to learn a *supervised* low-rank summary of the original exposure X that preserves the causal ERS, simultaneously achieving flexibility in estimation, interpretability of the causal effect, and valid causal inference.

BACKGROUND

We propose **causal sufficient dimension reduction (CSDR)** to address these challenges. Let Y^x be the potential outcome under X . The exposure-response surface (ERS) is

$$\mu(x) = \mathbb{E}(Y^x), \quad \mu : \mathbb{R}^p \rightarrow \mathbb{R}.$$

CSDR assumes the ERS depends on X only through a low-dimensional index,

$$\mu(x) = g(\beta^T x), \quad \beta \in \mathbb{R}^{p \times d}, \quad d < p,$$

for some smooth function $g(\cdot)$. The estimand is not a particular basis β , but the causal central mean subspace (CCMS) or $\mathcal{S}_{\mathbb{E}(Y^x)}$, the intersection of all $\text{span}(\beta)$ for matrices β that satisfy the dimension reduction constraint.

Nabi et al. (2022) proposed estimating $\mathcal{S}_{\mathbb{E}(Y^x)}$ with a locally efficient score based on marginal structural models [1]. It applies inverse propensity weighting and g -comp to the noncausal score for β . Estimation proceeds by Newton-Raphson numerical search routine.

Nuisance functions:

$$\begin{aligned} \mu_\beta(z) &:= \mathbb{E}(Y^z) && \text{(causal ERS),} \\ \nu_\beta(z) &:= \mathbb{E}\{\alpha(X)^z\} && \text{(causal exposure regression),} \\ \pi(X|C) &:= f_{X|C}(X|C) && \text{(generalized propensity score),} \\ m(X, C) &:= \mathbb{E}(Y|X, C) && \text{(outcome regression).} \end{aligned}$$

Locally efficient association-based score:

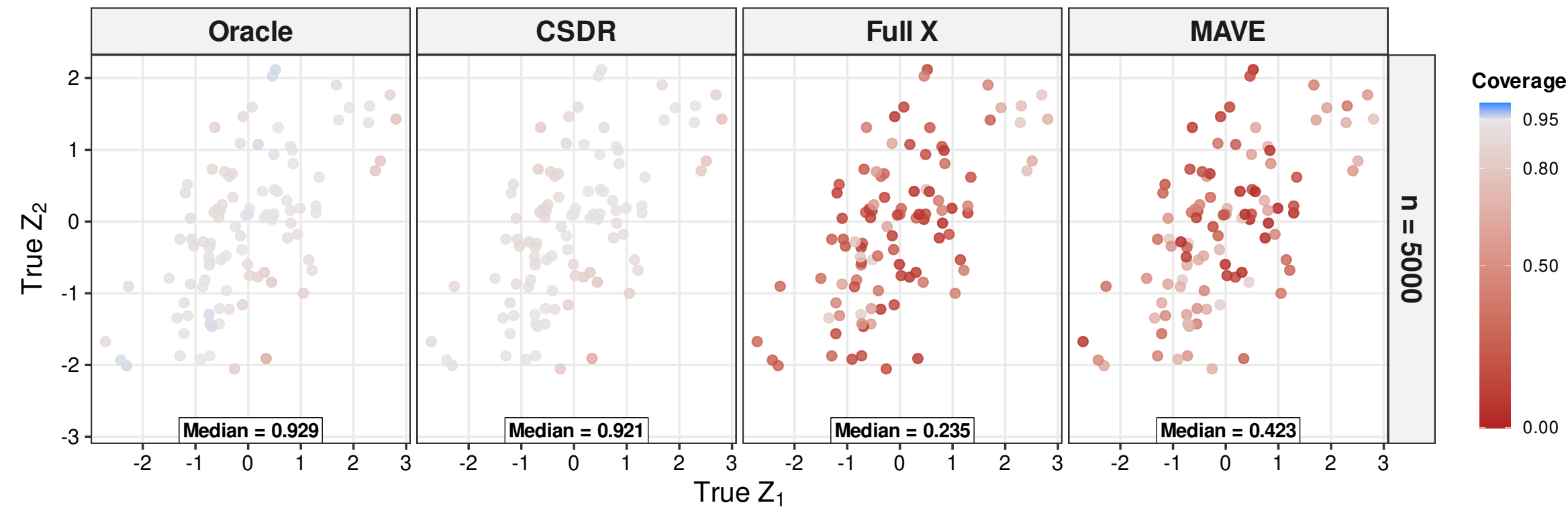
$$U(\beta) = \{Y - \mu_\beta(Z)\}\{\alpha(X) - \nu_\beta(Z)\}.$$

Locally efficient causal score, as a function of $U(\beta)$:

$$\tilde{U}(\beta) = \frac{\pi^*(X)}{\pi(X|C)} \{U(\beta) - \mathbb{E}[U(\beta)|X, C]\} + \mathbb{E}^*[\mathbb{E}[U(\beta)|X, C]|C].$$

This is challenging to implement due to 1) unclear choice of $\alpha(\cdot)$ and $\pi^*(\cdot)$ 2) high burden of β -dependent nuisance estimation. We propose CSDR as an alternative approach.

SIMULATION AND REAL DATA ANALYSIS



Simulations with $p = 10$ and $d = 2$ compare CSDR with the Oracle (true β is known), Full X (causal inference using the full X vector and no dimension reduction), and MAVE (association-based dimension reduction) approaches. CSDR recovered $\mathcal{S}_{\mathbb{E}(Y^x)}$, selected the correct dimension, and achieved 95% coverage with performance similar to the Oracle, and improved greatly over benchmark methods.

CAUSAL SUFFICIENT DIMENSION REDUCTION

Observed data and causal target

(Y_i, X_i, C_i) with $Y_i \in \mathbb{R}$, $X_i \in \mathbb{R}^p$, $C_i \in \mathbb{R}^q$

Goal: A $p \times d$ basis matrix β of the causal central mean subspace $\mathcal{S}_{\mathbb{E}(Y^x)}$ where $\mu(x) = \mathbb{E}(Y^x) = g(\beta^T x)$, and $d < p$

Stage 1: Construct $(\tilde{Y}, \tilde{X})$ so that $\mathbb{E}(\tilde{Y} | \tilde{X} = x) = \mu(x)$.

Valid options include the regression adjustment (RA) estimate of $\mu(x)$, the doubly robust estimator (DR) (not shown), pseudo-outcome (PO), and residualized pair (RP).

Variant	$\tilde{Y}_j$	$\tilde{X}_j$
RA	$\mu(X_j) = n^{-1} \sum_{i=1}^n \hat{m}(X_j, C_i)$	X_j
PO	$\frac{Y_j - \hat{m}(X_j, C_j)}{\hat{\pi}(X_j C_j)} \left(n^{-1} \sum_{i=1}^n \hat{\pi}(X_j C_i) \right) + n^{-1} \sum_{i=1}^n \hat{m}(X_j, C_i)$	X_j
RP	$Y_j - \hat{\mathbb{E}}(Y C_j)$	$X_j - \hat{\mathbb{E}}(X C_j)$

Stage 2: apply association-based SDR to $(\tilde{Y}, \tilde{X})$

A local linear approximation to $g(\beta^T x)$ recovers β where w_{ij} are kernel weights. Minimum average variance estimation (MAVE) efficiently solves for β [2].

$$\hat{\beta}_d = \arg \min_{\beta, \beta^T \beta = I_d} \sum_{i=1}^n \sum_{j=1}^n [\tilde{Y}_i - \{a_j + b_j^T \beta^T (\tilde{X}_i - \tilde{X}_j)\}]^2 w_{ij}^{(d)}.$$

We also estimate the true dimension d of the CCMS using a Nadaraya-Watson based estimator.

$$\hat{d} = \arg \min_{1 \leq d \leq p} CV(d), \quad CV(d) = \frac{1}{n} \sum_{j=1}^n \left\{ \tilde{Y}_j - \hat{\mathbb{E}}(\tilde{Y}_j | \hat{\beta}_d^T \tilde{X}_j) \right\}^2.$$

Subspace Importance Score (SIS)

We analyze each original variable's causal importance to $\mathcal{S}_{\mathbb{E}(Y^x)}$ using $\lambda_j := [\hat{P}]_{jj}$ where $\hat{P} := \hat{\beta}(\hat{\beta}^T \hat{\beta})^{-1} \hat{\beta}^T$.

- **Basis-invariant:** depends only on $\text{span}(\hat{\beta})$, not the particular basis β .
- **Bounded:** $0 \leq \lambda_j \leq 1$ by property of projection matrices.
- **Additive:** $\sum_{j=1}^p \lambda_j = d$, so variable importance is an additive decomposition of d .

Post-reduction causal ERS estimation

Define the reduced exposure $\tilde{Z} := \hat{\beta}^T X$ and use a continuous exposure causal estimator with Z replacing X [3].

$$\hat{\mu}_{\hat{\beta}, DR}(z) = \frac{1}{n} \sum_{i=1}^n \left[\frac{K_{h_Z}(\tilde{Z}_i - z)}{h_Z^2 \hat{\pi}_{\hat{\beta}}(\tilde{Z}_i | C_i)} \{Y_i - \hat{m}_{\hat{\beta}}(z, C_i)\} + \hat{m}_{\hat{\beta}}(z, C_i) \right].$$

Pointwise 95% CIs for the ERS are generated using the sample variance of the estimated influence function.

THEORY AND ASSUMPTIONS

Assumptions:

- **(Causal identification - standard)** (i) consistency $Y^x = Y^X$, (ii) conditional ignorability $Y^x \perp X | C$, and (iii) strong positivity $\pi(x|c) \geq \epsilon > 0$ on the support. These ensure the causal ERS $\mu(x) = \mathbb{E}(Y^x)$ is identifiable.
- **(Causal identification - smoothness):** (i) $m(x, c)$ and $\pi(x|c)$ have bounded, continuous third derivatives with respect to x . (ii) $\text{Var}(Y|X, C)$ and its derivatives w.r.t. X are bounded uniformly. These are needed for identification of continuous exposures.
- **(Dimension reduction structure):** (i) the ERS can be written as $g(\beta^T x)$ for some $d < p$ (ii) the reduced exposure $Z = \beta^T X$ is well-defined and inherits positivity and conditional ignorability.
- **(Smoothness for SDR):** (i) marginal density of $f_X(\cdot)$ has bounded derivatives up to order 4 and compact support, (ii) $g(\cdot)$ has bounded continuous derivatives up to order 3, (iii) kernels are regular, and (iv) the first-stage estimator of $\tilde{Y}$ set to $\mu(X)$ satisfies a uniform rate $R_n = \sup_i |\hat{\mu}(X_i) - \mu(X_i)| = o(h^2)$.

Identification. The causal ERS is identified by:

$$\mu(x) = \mathbb{E}\{m(x, C)\} = \lim_{h \rightarrow 0} \mathbb{E} \left[\frac{1}{h^p} \frac{K_h(X - x)}{\pi(X|C)} Y \right].$$

For a reduced exposure Z , the same identification holds on the lower-dimensional support:

$$\mu_\beta(z) = \mathbb{E}\{m_\beta(z, C)\} = \lim_{h \rightarrow 0} \mathbb{E} \left[\frac{1}{h^d} \frac{K_h(Z - z)}{\pi_\beta(Z|C)} Y \right].$$

Theorem 1 (Convergence rate of $\hat{\beta}$). Let β span the true CCMS and $\hat{\beta}$ be the csMAVE estimate based on the ERS pair. If δ_n is $\{\log n / (nh^p)\}^{1/2}$, then

$$\|(\hat{\beta} - \beta) \beta^T\| = O_p(h^3 + h\delta_n + h^{-1}R_n).$$

Compare to the original MAVE convergence rate which is

$$O_p(h^3 + h\delta_n + h^{-1}\delta_n^2).$$

The first two terms are standard MAVE error; the $h^{-1}R_n$ term replaces the old term and captures the impact of estimation from the first stage.

Theorem 2 (Consistency of $\hat{d}$). If the first-stage ERS estimator satisfies $R_n = o_p(h^2)$, then the MAVE dimension selector remains consistent:

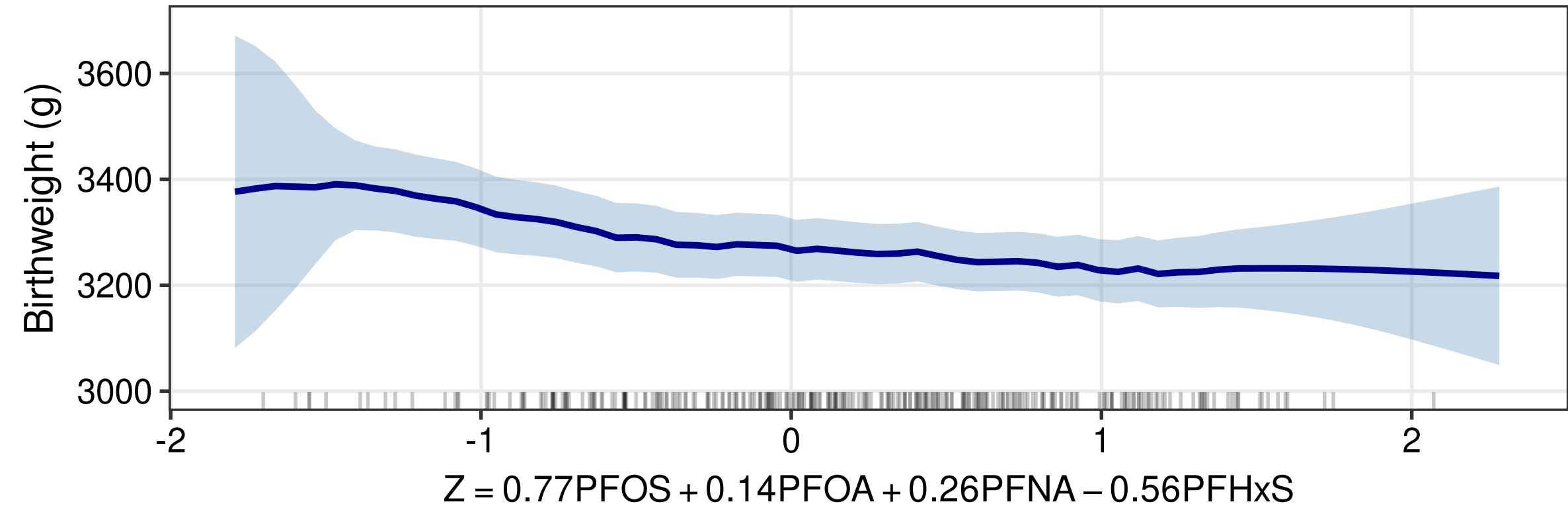
$$\Pr(\hat{d} = d_0) \rightarrow 1.$$

KEY POINTS

- **Computational efficiency:** we developed a two-stage estimator for the CCMS that reduces the computational burden of the previous MSM approach.
- **Recovers the causal subspace:** The method consistently estimates the CCMS and its dimension d with convergence rate dependent on how well $(\tilde{Y}, \tilde{X})$ are estimated.
- **Modular and flexible:** any valid first-stage causal estimator (RA, DR, PO, RP) can be combined with association-based SDR methods like MAVE, enabling robustness and adaptability through use of existing tools.
- **Dimension reduction improves inference:** projecting to $Z = \beta^T X$ reduces the smoothing dimension from p to d , improving nuisance estimation, efficiency and inference.
- **Variable importance:** We proposed a basis invariant measure λ_j that gives a transparent and interpretable measure of variable importance for the causal ERS.

References:

- [1] Nabi et al. (2022). Semiparametric causal sufficient dimension reduction of multidimensional treatments. UAI.
- [2] Xia et al. (2002). An Adaptive Estimation of Dimension Reduction Space. JRSS-B.
- [3] Colangelo, K. and Lee, Y.-Y. (2025). Double Debiased Machine Learning Nonparametric Inference with Continuous Treatments. JBES.



We applied CSDR to a cohort of 305 African-American mothers to estimate the joint effect of four PFAS exposures on infant birthweight. We estimated a 1D PFAS mixture with $\hat{Z} = 0.77 \text{PFOS} + 0.14 \text{PFOA} + 0.26 \text{PFNA} - 0.56 \text{PFHxS}$. Birthweight decreased with increasing PFAS exposure, and the SIS values $\lambda = (0.60, 0.02, 0.07, 0.31)$ identified PFOS as the dominant contributor.